

Weighted Graph-Structured Lambda Theories

Stochastic and quantum execution for MeTTaIL, with a spiking recurrent network as a worked example

A F1R3FLY.io working note

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Abstract

We equip the rewrite rules of a finitely presentable graph-structured lambda theory (GSLT), as presented in MeTTaIL, with *weight maps*: finite maps whose keys are spatial-behavioural formulae refining the left-hand side of a rule and whose values are weights. A weight map instance is part of a state specification, so a rewrite may update it; weights are therefore dynamic. We develop the construction in three movements. First, we impose a *partition discipline* on the key set — the refinements of a rule’s left-hand side must be pairwise exclusive and jointly exhaustive — and show that this single requirement is what makes the weighting a total, single-valued classification of redexes; we then say what it costs, which is a dilemma rather than a theorem: a partition stated inside the generated logic demands Boolean structure of the target specification, and a partition stated outside it forfeits specification closure. Second, with weights in $\mathbb{R}_{\geq 0}$ we define propensity as $\mathbf{a} = w(\varphi) \cdot \sum_k g(k)$ over classified, funded redex positions and obtain a Gillespie simulator; we prove that the construction restricted to the summation-free π -calculus, static maps, and channel-identity keys is exactly the Stochastic Pi Machine, and that dynamic maps take it strictly beyond. Third, with weights in $\mathbb{C}$ we use the complex coalgebra coefficients directly as amplitudes of a linear transition operator on the Hilbert space spanned by reachable *configurations*, terms paired with weight maps. A single global normalisation makes that operator a contraction T without changing its relative phases or its normalised postselected output. The Julia–Halmos defect construction embeds T in a unitary on the system and one auxiliary qubit: the zero branch applies T and the one branch records failure. Repeating the construction with n fresh auxiliaries and postselecting their all-zero state implements T^n . Absolute squares enter only when those auxiliaries or the system are measured, so interference occurs in the rewrite amplitudes themselves; structural equations remain quotient identifications and no separate Hamiltonian is introduced. The worked example is an asynchronous stochastic spiking recurrent network in a weighted rho theory: neurons are persistent for-comprehensions, synaptic efficacies live in the weight map rather than in the payload, thresholds are join structure, a saturating response emerges from the chain rather than being written into a term, and a minimal local Hebbian potentiation rule is exactly the refinement entries’ update functions — so that learning and inference are the same transition relation, differing only in which component of the state they touch.

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1 Introduction

A rewrite rule’s left-hand side is a type. Terms inhabit it; the rule fires at its inhabitants. This note is about what happens when one refines that type and attaches a number to each refinement.

The move is small to state and consequential to develop. A GSLT is a triple (Σ, E, R) : a signature which is in general a lambda theory, a set of equations imposing a structural congruence, and an adjoined graph of rewrite rules [?]. Given such a presentation, the OSLF family of algorithms generates a logic from it, with one structural connective per term former of Σ and one context-labelled modality per rule of R together with a choice of redex position [?, ?]. The formulae of that generated logic are exactly the vocabulary in which one can refine a rule’s left-hand side — and they are generated, not designed, so the refinements come for free with the theory. To weight a rule is to give a finite map

$$\varphi_1 \mapsto v_1, \quad \dots, \quad \varphi_m \mapsto v_m$$

from such refinements to weights. A term’s redexes are then classified by which refinement they satisfy. Real weights turn the enabled redexes into a rate distribution; complex weights turn the same classified transition graph into a linear amplitude operator.

Two things distinguish this from the existing literature on stochastic process calculi.

Keys are formulae, not names. In Priami’s stochastic π -calculus [?] and in Phillips and Cardelli’s Stochastic Pi Machine (SPiM) [?, ?], a rate is attached to a *channel*. That is one refinement among many: “this redex is a communication on n ” is a particular formula in the generated logic, and there is no reason the modeller should be restricted to it. Once keys are formulae, a rate may depend on the shape of the payload, on the *namespace* a channel belongs to — which in a reflective calculus is real structure, reachable by the classifier through the name predicate $@\varphi$, and is what lets one key cover a growing population of channels (§??) — or on any Boolean combination of these. Channel identity is recovered as the degenerate case (§??). We are careful not to claim more than that. The classifier of Proposition ?? evaluates its formula at the matched left-hand side $L\sigma$ and not at $k[L\sigma]$, so the ambient context is not visible to a key at all; properties of a redex’s *position* are the business of the geometric factor $g(k)$ (§??), and the division of labour between the two is clean only because it is respected.

Maps are state, not declaration. A weight map instance is part of a state specification. A rule’s refinement entries therefore carry not only a weight but an *update function* on the whole map, applied when the rule fires. Rates change during execution. SPiM does not permit this — its rates are fixed at declaration — and the restriction is not incidental to its design but constitutive of it, since a static rate function is what makes the underlying object a time-homogeneous CTMC over terms alone. We recover Markovianity by a different route: the weight map is *in* the state, so the chain is over configurations (P, w) rather than over terms (§??). This is the technical content of “the map may be updated,” and it is what the worked example of §?? needs, because plasticity is nothing other than a weight-map update.

Two further points of scope. The construction is generic over the GSLT: it applies to any theory one can write down in MeTTaIL’s **language!** block [?], so one obtains a stochastic machine for the ambient calculus, for a user-defined DSL, or for the ρ calculus, by writing the theory rather than by writing a simulator. And the weight codomain selects an execution. We treat two instances: $\mathbb{R}_{\geq 0}$, giving classical Gillespie simulation (§??); and $\mathbb{C}$, giving a complex transition operator together with a postselected unitary dilation (§??). The two instances share classification and the weighted transition graph but not a propensity interpretation. A third instance — weights in a quantale, giving a fuzzy calculus — is deliberately out of scope here; see [?].

What is new. Beyond the two distinguishing points above:

- (i) The *partition discipline* of §??, and the observation that it is not a technicality but the condition under which weighting is well defined at all — together with an account of what it costs, namely that stating it internally demands Boolean structure of the target logic specification while stating it externally forfeits specification closure (§??).
- (ii) A propensity formula (§??) that separates four factors which the literature usually conflates: the refinement’s weight, a geometric factor accumulated up the context rules, redex multiplicity, and a funding gate supplied by cost accounting.
- (iii) Theorem ??: the summation-free fragment of SPiM is the construction’s restriction, exactly.
- (iv) Definition ??, which uses the complex weights before any Born rule: derivations with a common target add as amplitudes in one matrix element, so their relative phases can reinforce or cancel.
- (v) The unitary defect dilation of Theorem ?? and the all-zero branch theorem (Theorem ??), which implement T with one auxiliary qubit and T^n with n fresh auxiliaries.
- (vi) The observation (§??) that the OSLF-generated formulae give canonical basis-diagonal projective measurements of the postselected state.
- (vii) The spiking-network example of §??, whose thesis is that *synaptic efficacy is represented as transition intensity in the operational semantics, and the same transition updates that intensity*: the efficacies live in the weight map, the threshold lives in the join structure of a for-comprehension, a saturating response is emergent, and learning is a weight-map update performed by the same rewrite that performs inference.

Where this note sits. It supplies the construction that [?] cites as *Stochastic simulation for decorated ciGSLTs* and fills the placeholder section of [?]. It depends on [?] for GSLTs and OSLF, on [?] and [?] for the shape of the generated spatial-behavioural logic — which is spelled out in those papers and not yet present in the implementation branch — and on [?] for cost accounting, which enters only in §?? and only as a gate.

Revision note. This is the second version of this note. The first was read by an external referee, and most of what follows is a response to that reading. Where a claim has been withdrawn or weakened we say so at the point of change rather than silently, since a reader of the first version deserves to be told which of its statements not to rely on. The quantum construction is now built over a basis of configurations and uses complex weights directly as entries of a transition operator. The former magnitude-square interpretation, Lindblad jump construction, and Hamiltonian on un-quotiented structural equations have been removed; they discarded the phases before the rewrite dynamics and treated representatives of one structural class as distinct states. The replacement normalises the assembled operator globally, dilates it unitarily by its defects, and obtains powers by all-zero postselection (§??). The other substantive changes remain: the necessity claim about Boolean target specifications is weakened and replaced by a dilemma (§??); the locality lemma is restated for the fragment in which it is true, with a matching requirement added to admissibility (§??); the emergent neural response is exponential-saturating and not logistic (§??); the SPiM theorem is titled for the fragment it actually proves (§??); the CTMC generator’s diagonal is corrected for self-transitions (Remark ??); and §?? is rewritten against the implementation that exists rather than against the prototype it superseded.

2 Background, compressed

We recall only what the construction uses, and forward everything else to [?].

2.1 GSLTs and their presentations

A *graph-structured lambda theory* is a triple (Σ, E, R) . The signature Σ is a lambda theory — terms may contain variables, and binding is part of the presentation, which is what distinguishes the setting from graph-enriched Lawvere theories. E is a set of equations imposing a structural congruence $\equiv$ on terms. R is a graph of rewrite rules adjoined to the structure as a Lawvere theory of graphs, with source and target maps satisfying the minimal requisite equations. The adjective *graph-structured* captures the rewrites; *lambda theory* captures term languages with variables.

Rules come in two kinds, and the distinction is load-bearing throughout this note.

Definition 1 (Base and context rules). A *base rule* is hypothesis-free:

$$r. \vdash L \rightsquigarrow R,$$

with L, R terms of some sort τ over Σ , possibly with pattern variables. A *context rule* is conditional on a rewrite:

$$c. \mid S \rightsquigarrow T \vdash C[S] \rightsquigarrow C[T],$$

for a one-hole context former C . Base rules are the ground of the dynamics; context rules are *addressing* — they say where a base rule may fire, not that anything happens.

This is MeTTaIL’s actual surface syntax for its **rewrites** block [?]. In the **rho** presentation, **Exec** and the communication rule are base; **ParCong** and **NewCong** are context rules.

An *interactive* GSLT is one whose base rules’ left-hand sides are headed by a distinguished family of formers K — the interaction cut — across which program and environment meet: application for the lambda calculus, parallel composition for CCS, π and **rho**. A *continued* interactive GSLT (ciGSLT) additionally has surfaces and continuations. Everything below is stated for ciGSLTs, though only the interactive structure is used.

2.2 Locations

Write $\partial\mathcal{T}$ for the space of one-hole contexts, in the McBride-derivative sense: a location is a pair $(\partial T, T)$ of a context and the term type. We use $\partial\mathcal{T}$ as a single object playing three roles, as in [?]: the index set of the generated modalities, the labels on the transition system, and the positions inside a term. It is this identification that lets a weight be attached to a *position* and read as a modality.

Definition 2 (Redex). Let $r. \vdash L \rightsquigarrow R$ be a base rule and P a term. A *redex* of r in P is a *selection* of a one-hole context $k \in \partial\mathcal{T}$ and a substitution σ for the pattern variables of L from the decomposition of P , such that $P \equiv k[L\sigma]$. Two redexes are identified when they select the same occurrences; interchanging two occurrences of $\equiv$ -equal subterms yields *distinct* redexes, which happen to have $\equiv$ -equal contracta. Write $\text{Red}_r(P)$ for the set of redexes of r in P , and $\text{Red}(P) = \coprod_r \text{Red}_r(P)$.

$\text{Red}_r(P)$ is finite for finite P and finitely many rules. The convention on identification is Gillespie’s, and it is the one that matters: two molecules of the same species offer two ways for a reaction to occur, not one, and a stochastic simulator that identifies them runs at half speed. In an associative–commutative setting the reactants are drawn from a multiset, so $|\text{Red}_r(P)|$ is a count of selections from a bag — In · Out for a binary interaction drawing one

reactant from each of two species, $\binom{n}{2}$ for one drawing two from the same species. Structural congruence is used to *find* the decompositions, not to collapse them. Remark ?? records what happens when distinct redexes do share a contractum.

2.3 The generated logic

The following is generated by OSLF from the presentation and is not ours to design. We state it for the rho calculus — term formers $\mathbf{0}$, $*n$, $n!(Q)$, $\mathbf{for}(x \leftarrow n)P$, parallel composition, and $@P$, with parallel composition associative–commutative and a single communication rewrite — and note that everything transfers verbatim to any other presentation, with the connective list changing to match the formers. The presentation follows [?, §2] and [?, §3]; it is not present in the implementation branch, whose logic is still under development.

Structural connectives. One connective per term former. $\mathbf{out}(n)\varphi$ holds of $n!(Q)$ with $Q \models \varphi$; $\mathbf{in}(n)\varphi$ is the analogue on receipts. Because parallel composition is associative–commutative, the generated connective for it is a separating conjunction:

$$P \models \varphi \mid \psi \iff P \equiv Q \mid R \text{ with } Q \models \varphi, R \models \psi.$$

Behavioural modalities. One primitive modality $\langle K_j \rangle$ per rewrite rule *and choice of redex position*, labelled by the one-hole context in which the step fires. The index is not decoration. For rho, with a single interaction rule, all the information in a label is in the position: for $P \equiv K_j[\mathbf{for}(x \leftarrow n)Q \mid n!(Q')]$ the label K_j is everything else. Thus $\exists j. \langle K_j \rangle \varphi$ and $\forall j. [K_j] \varphi$ quantify over redex positions, and a bare $\langle K \rangle$ abbreviates the existential.

Name predicates. A name is a quoted process, so

$$n \models @ \varphi \iff n = @P \text{ with } P \models \varphi,$$

which is namespace logic [?]: a namespace is described by a property rather than enumerated. This matters in §??, where synapses are picked out by structured names.

Propositional layer. Conjunction and $\top$ come from the finite-limits fragment; disjunction, implication and negation require more, and a target specification supplies them or does not [?]. §?? shows that weighting *forces* the choice.

Additions. A greatest fixed point $\nu X. \varphi$ with X positive; the hidden-name quantifier $\mathbf{H}x. \varphi$; graded separation $\varphi \mid_k \psi$ [?, §2.3]. We use ν only in §??.

Definition 3 (Depth). The depth $d(\varphi)$ of a structural formula is the maximal nesting of term-former connectives and prefixes: $d(\top) = 0$, $d(\mathbf{out}(n)\varphi) = d(\mathbf{in}(n)\varphi) = 1 + d(\varphi)$, $d(\varphi \mid \psi) = \max(d(\varphi), d(\psi))$, $d(@\varphi) = d(\varphi)$, $d(\mathbf{H}x. \varphi) = 1 + d(\varphi)$.

2.4 Ideal logic, checkable fragment

One methodological commitment governs §??. Following [?, §1.3], we distinguish the *ideal* logic, adequate for the idealised calculus and the thing that *defines* a semantic notion, from *fragments*, restricted so that model checking terminates, which *decide* it. A logic whose model checking terminates is necessarily not adequate for the whole of a Turing-complete host. The relevance here is sharp and appears at Requirement ???: a simulator evaluates its key formulae at every step, so keys must come from a checkable fragment. What is elsewhere a methodological trade-off is here an executability constraint.

3 Refining a left-hand side

3.1 The keys

Fix a base rule $r. \vdash L \rightsquigarrow R$ with L of sort τ . The set of redexes of r is carved out by the shape of L ; we want to subdivide it. Write $L^\sharp$ for the structural formula that characterises the left-hand side pattern — for the **rho** communication rule, $L^\sharp = \text{in}(n)\top \mid \text{out}(n)\top$ with n existentially bound. The candidate keys are the formulae of the generated logic that *refine* $L^\sharp$.

Definition 4 (Refinement). A formula φ *refines* $L^\sharp$, written $\varphi \sqsubseteq L^\sharp$, when $\models \varphi \Rightarrow L^\sharp$. The refinements of $L^\sharp$, ordered by entailment, form the *refinement lattice* $\mathcal{R}(r)$ of the rule.

We now impose three requirements on a key set $\Phi_r \subseteq \mathcal{R}(r)$. The first is the substantial one.

Definition 5 (Partition discipline). $\Phi_r = \{\varphi_1, \dots, \varphi_m\}$ is a *partition* of r when

(P1) *Exclusivity*: $\models \neg(\varphi_i \wedge \varphi_j)$ for all $i \neq j$;

(P2) *Exhaustiveness*: $\models L^\sharp \Rightarrow \bigvee_i \varphi_i$.

Definition 6 (Requirements on a key set). A key set is *admissible* when it is a partition and, in addition:

(R1) *Decidability*: each φ_i lies in a fragment for which model checking terminates on the terms of interest.

(R2) *Locality*: $d(\varphi_i) \leq \alpha_r < \infty$ for all i , for some bound α_r fixed with the rule.

(R3) *Structurality*: each φ_i lies in the structural fragment — **out**, **in**, **@**, **|**, ground predicates and the propositional connectives — carrying no behavioural modality $\langle K_j \rangle$, no fixed point $\nu X.\varphi$ and no hidden-name quantifier.

?? is forced by execution: a simulator must classify every enabled redex before it can sample, so a key whose evaluation may diverge is a key that stops the simulation. ?? and ?? are forced by *efficient* execution and pay off together in Lemma ???. ?? is new in this version, and it is a restriction on *keys* alone: the behavioural modalities and the fixed point remain available for stating and checking properties of the resulting chain, and §?? uses ν for exactly that. What they may not do is determine a rate. A rate that depends on what a redex can do next cannot be recomputed locally when something elsewhere in the term changes, and the incremental scheme of §?? is the thing that would break.

3.2 What the partition buys

Proposition 7 (Classification). *Let Φ_r be a partition of r . Then the assignment*

$$\text{cls}_r : \text{Red}_r(P) \longrightarrow \Phi_r, \quad \text{cls}_r(k, \sigma) = \text{the unique } \varphi_i \text{ with } L\sigma \models \varphi_i$$

is a total function, for every term P .

Proof. $(k, \sigma) \in \text{Red}_r(P)$ gives $P \equiv k[L\sigma]$, so $L\sigma \models L^\sharp$ by construction of $L^\sharp$. Exhaustiveness gives $L\sigma \models \varphi_i$ for some i , which is totality; exclusivity gives at most one such i , which is single-valuedness. Well-definedness on $\equiv$ -classes holds because satisfaction is $\equiv$ -invariant, the generated connectives being generated from Σ modulo E . $\square$

This is exactly what fails without the discipline, and it fails in two directions. Without exhaustiveness the weighting is partial: some enabled redexes have no rate, and one must invent a convention (fire them at rate zero? at a default rate? refuse to run?) that is not part of the specification. Without exclusivity the weighting is multi-valued, and one must invent an *aggregation* (take the most specific key, if a unique minimum exists; sum the weights of all satisfied keys; take the maximum). Each convention is defensible and each yields a different simulator from the same specification, which is precisely the situation a specification exists to prevent.

Remark 8 (Completing a partition is cheap). The discipline is less onerous than it looks. Given any pairwise-exclusive family $\{\varphi_1, \dots, \varphi_m\}$, the family $\{\varphi_1, \dots, \varphi_m, \varphi_\perp\}$ with $\varphi_\perp := L^\# \wedge \neg \bigvee_i \varphi_i$ is a partition, and a modeller who wants the unlisted redexes inert simply sets $w(\varphi_\perp) = 0$. Exclusivity is the real constraint; exhaustiveness is a completion. In practice, keys of the form “the redex is a communication on channel n ,” indexed by distinct n , are exclusive for free.

3.3 What stating the partition costs

Proposition 9 (Boolean demand, for an internal statement). *Suppose the partition conditions (P1) and (P2) are to be stated in the generated logic itself, as Definition ?? writes them. Then that logic must carry negation, conjunction and disjunction, and a target logic specification supplying only the finite-limits fragment will not do.*

Proof. Immediate from the shape of (P1) and (P2): (P1) mentions $\neg$ and $\wedge$, (P2) mentions $\Rightarrow$ and $\bigvee$. Finite limits supply $\wedge$ and $\top$ only [?]. $\square$

The hypothesis is doing all the work, and the first version of this note dropped it, concluding instead that *one cannot weight a theory whose generated logic is too weak to be adequate for it*. That does not follow. A key family drawn from a positive fragment may perfectly well be pairwise disjoint and jointly exhaustive; what a weak logic prevents is not the *holding* of (P1) and (P2) but their *expression* as entailments between formulae of the language the keys are written in. Exclusivity and exhaustiveness can instead be discharged in the metatheory — by an elaborator that checks them on the instances that arise, or by a proof about a schema of keys — and the weighting is then perfectly well defined. The choice is a genuine one, and both branches cost something.

Observation 10 (The cost of an external partition). If (P1) and (P2) are discharged in the metatheory rather than in the generated logic, the specification is no longer *closed*: whether a key set is admissible becomes a judgment the object language cannot express, so two implementations conforming to the same written specification may disagree about admissibility, hence — by Proposition ?? — about the classification, hence about the sampled distribution. That is precisely the failure the partition discipline was introduced to prevent (§??), relocated one level up rather than removed.

So a weighted GSLT must choose between a Boolean target specification and an external admissibility judgment, and it should say which. Our own implementation takes the second branch and pays the stated price: its partition check is a procedure over witnesses, sound as a refuter and silent as a proof (§??).

On the first branch there is still a convergence worth recording. The omnibus identifies possession of something like a negation and a conjunction as exactly the condition under which a generated logic enjoys an adequacy theorem, and notes that Hypercube-style members of the OSLF family, which generate type systems, offer no such thing [?]. Proposition ?? says that an internal statement of the partition makes the same demand for an independent

reason. A weighting is a claim about which behaviours are distinguishable enough to be priced differently, and if that claim is to be made inside the generated logic it needs the expressive power of the claim that they are distinguishable at all.

Requirement ?? then pulls the other way, since a terminating fragment is necessarily inadequate for a Turing-complete host. The two are not in conflict, because they apply to different artifacts: adequacy is a property of the ideal logic in which the specification is *stated*, and termination is a property of the fragment in which the keys are *evaluated*. A modeller writes a partition in the ideal logic, then checks that the particular keys chosen happen to lie in a checkable fragment. §?? exhibits keys of depth 2, for which the check is trivial.

3.4 Locality

Lemma ?? below is what licenses an incremental simulator. It is true of the structural fragment and false of the generated logic as a whole, and the first version of this note stated it for the whole. The counterexamples are not exotic: a behavioural modality $\langle K_j \rangle \varphi$ inspects the successors of a term rather than a bounded syntactic neighbourhood of a position, and a greatest fixed point may be refuted only by unbounded inspection, so neither is confined by a depth bound in the sense Definition ?? measures. ?? is what restricts the lemma to the fragment where it holds.

A second correction concerns the measure. “Distance in the term tree” is not well defined modulo an associative–commutative equation, since a parallel composition is a multiset and has no canonical tree; a lemma whose radius depends on a choice of tree is a lemma about a presentation. We use instead the *component distance* $\delta(k, k')$ between two positions: the number of term-former occurrences on the shortest path joining them in a decomposition witnessing both, counting one parallel composition as a single former whatever its arity. δ is $\equiv$ -invariant, because the decompositions it quantifies over are exactly those the generated connectives quantify over, and it agrees with tree distance in a presentation with no AC formers.

Lemma 11 (Locality of reclassification, structural fragment). *Let all keys of all rules satisfy ?? and ??, with depth at most α . Let $P \equiv k[L\sigma]$ and let $P' \equiv k[R\sigma]$ result from firing that redex. Then for any redex (k', σ') of any rule surviving in P' with $\delta(k, k') > \alpha + |L| + |R|$, the class $\text{cls}(k', \sigma')$ is unchanged.*

Proof. By induction on a structural formula φ with $d(\varphi) \leq \alpha$, satisfaction of φ at a position is determined by the components within α nested formers of it: **out**, **in** and **@** descend one former; $\wedge$, $\vee$ and $\neg$ descend none; and the separating conjunction quantifies over decompositions of the subterm at that position rather than of the whole, so it stays inside the same neighbourhood. No case appeals to a rewrite or to an unbounded unfolding, which is where ?? is used and where the lemma would fail without it. The rewrite replaces $L\sigma$ by $R\sigma$ and leaves k pointwise fixed, so a neighbourhood of radius α disjoint from the replaced material sees an identical multiset of components; the offset $|L| + |R|$ accounts for neighbourhoods overlapping the replaced region. $\square$

The point is algorithmic. Recomputing cls over the whole term at every step makes the simulator quadratic in term size; Lemma ?? licenses an incremental propensity update in the style of Gibson and Bruck’s next-reaction method [?], touching only a bounded neighbourhood of the fired redex. This is where ?? and ?? earn their place, and it is the reason to prefer shallow structural keys where the modelling permits. A modeller who wants a modal key is not forbidden one — the logic is restricted for keying, not for checking — but the incremental scheme is then unavailable and the simulator must reclassify globally, which is what our implementation does and what §?? records.

4 Weighted rewrite systems

4.1 The weight codomain

Remark 12 (Two codomains, two operational meanings). Weights are drawn from a commutative semiring $\mathcal{K}$, and we treat two instances,

$$\mathcal{K} = \mathbb{R}_{\geq 0} = (\mathbb{R}_{\geq 0}, +, \cdot, 0, 1) \quad \text{and} \quad \mathcal{K} = \mathbb{C}.$$

They share the key sets, classification, updates, and weighted transition graph, but not the final interpretation of a coefficient. In $\mathbb{R}_{\geq 0}$ a weight is a rate, with units of inverse time, and Definition ?? assembles those rates for Gillespie simulation. In $\mathbb{C}$ a weight is an amplitude. It is inserted directly into the matrix element of Definition ??; there is no map $z \mapsto |z|^2$ before that operator is formed. Magnitude squares appear only as Born probabilities after the operator has been embedded in a unitary and an auxiliary qubit has been measured. Thus the complex instance retains phase at the point where different derivations are added.

Remark 13 (Rates and global contraction normalisation). We take the real codomain to be $\mathbb{R}_{\geq 0}$ rather than the unit interval. A real weight is a rate and there is nothing to normalise it against. The unit interval is available as a derived presentation: fixing a global clock rate $\Lambda \geq \mathbf{a}_0$ and setting $p_i = \mathbf{a}_i/\Lambda$ gives the uniformised discrete-time chain. Uniformisation is a useful implementation strategy and a poor local specification, because dynamic maps make the required global bound a reachability obligation.

The complex construction likewise imposes no entrywise bound such as $|z| \leq 1$. Contractivity is a property of the assembled matrix, not of each weight separately: coherent sums can make an operator large even when every entry of the map lies in the unit disk. Definition ?? therefore forms the full amplitude operator T_0 and divides once by $\alpha = \max\{1, \|T_0\|\}$. This preserves every relative phase and changes only the success probability of the postselection, not its normalised conditional output.

4.2 Weight maps and augmented rules

Definition 14 (Weight map). Let (Σ, E, R) be a GSLT with admissible key sets Φ_r for each base rule r . A *weight map* is a function

$$w : \coprod_r \Phi_r \longrightarrow \mathcal{K}.$$

Write $\mathcal{W}$ for the set of weight maps. Since each Φ_r is finite and there are finitely many rules, $\mathcal{W} \cong \mathcal{K}^N$ for a fixed N .

Definition 15 (Configuration). A *configuration* is a pair $\mathbf{c} = (P, w)$ with P a term modulo $\equiv$ and $w \in \mathcal{W}$. Write $\mathcal{C}$ for the set of configurations.

The phrase “a map instance is part of a state specification” is Definition ??, and everything downstream turns on it.

Definition 16 (Augmented rules). An *augmented base rule* is

$$r. \vdash L \rightsquigarrow R \ [w_r] \ \{ \varphi_1 \Rightarrow (v_1, u_1), \dots, \varphi_m \Rightarrow (v_m, u_m) \}$$

where $\{\varphi_i\}$ is an admissible key set for r , $v_i \in \mathcal{K}$ are the initial weights, and $u_i : \mathcal{W} \rightarrow \mathcal{W}$ are *update functions* on the whole map. An *augmented context rule* is

$$c. \mid S \rightsquigarrow T \vdash C[S] \rightsquigarrow C[T] \ [g_c] \ \{ f_c \}$$

with $g_c \in \mathbb{R}_{\geq 0}$ a *geometric factor* and $f_c : \mathcal{W}^* \times \mathcal{W} \rightarrow \mathcal{W}$ a *fold function*.

The initial weight map is $w_0(\varphi_i) = v_i$; the leading $[w_r]$ is a per-rule scalar, convenient in practice and formally absorbable into the v_i , which is how we treat it below.

4.3 Why context rules carry a factor and not a propensity

The design decision that most distinguishes Definition ?? from the obvious alternative is that context rules do not have propensities of their own.

The obvious alternative gives every rule, base and context alike, a weight and lets the simulator select among all of them. It double-counts. A step of the transition system is a base rule firing *somewhere*; the context rules composing the path from the root to that somewhere are the address of the firing, not additional events. In the vocabulary of the energetics reading of OSLF-decorated GSLTs, structural rules are addressing, not dynamics, and incur no primitive cost. Selecting a context rule and then a base rule as though they were independent draws makes a single event into two.

So context rules contribute multiplicatively, to the propensity of the base firing they address:

Definition 17 (Geometric factor). Let $k \in \partial\mathcal{T}$ be a position, obtained by composing context rules $c_1, \dots, c_p$ from the root inwards. Its *geometric factor* is $g(k) = \prod_{q=1}^p g_{c_q} \in \mathbb{R}_{\geq 0}$.

Setting every $g_c = 1$ recovers “addressing is free” exactly, and is the default we recommend. The reason to permit $g_c \neq 1$ is that it is the natural home for a *spatial* bias: g is a weighting on the reduction graph’s edges that depends only on where in the term the interaction surfaces are, as against $w(\varphi)$, which depends on what is being transferred. This is the geometric-versus-matter split of the gravitation correspondence [?] appearing here as an implementation detail, and we flag it as such without pursuing it.

4.4 Dynamics

Definition 18 (Weighted transition). Fix a configuration $\mathfrak{c} = (P, w)$. Choose $(k, \sigma) \in \text{Red}_r(P)$ with $\text{cls}_r(k, \sigma) = \varphi_i$, where the position k is composed of context rules $c_1, \dots, c_p$. Then $\mathfrak{c}$ steps to $\mathfrak{c}' = (P', w')$ where

$$\begin{aligned} P' &\equiv k[R\sigma], \\ w' &= f_{c_1}(\vec{\omega}_1, f_{c_2}(\vec{\omega}_2, \dots, f_{c_p}(\vec{\omega}_p, u_i(w)) \dots)). \end{aligned}$$

the fold functions being applied from the redex outwards, with $\vec{\omega}_q$ the maps arising from the sibling subterms at level q .

In the common case where every f_c is the projection onto its second argument — “the context does not modify the map” — this collapses to $w' = u_i(w)$, and we write it so below.

Definition 19 (The coalgebra). Let $\mathcal{K}^{(-)}$ denote the free- $\mathcal{K}$ -semimodule monad, $\mathcal{K}^{(X)}$ being finitely supported functions $X \rightarrow \mathcal{K}$. A weighted GSLT determines a coalgebra

$$\gamma : \mathcal{C} \longrightarrow \mathcal{K}^{(\mathcal{C})}, \quad \gamma(\mathfrak{c})(\mathfrak{c}') = \sum_{\substack{(r, k, \sigma) \\ \text{stepping } \mathfrak{c} \rightarrow \mathfrak{c}'}} w(\text{cls}_r(k, \sigma)) \cdot g(k).$$

This is the target of the whole construction: a decorated ciGSLT plus an adequate spatial-behavioural logic plus a weight-map decoration gives a $\mathcal{K}$ -weighted coalgebra. Assembling the decorations of a fixed theory into a category and taking the Grothendieck construction gives the fibred form; the simulator is a functor out of it. We do not need the fibred statement below and record it only to fix the picture.

Remark 20 (Sum over derivations, not over targets). The sum in Definition ?? runs over *derivations* that reach $\mathfrak{c}'$, not merely over the target. In $\mathbb{R}_{\geq 0}$ this is the familiar statement that parallel routes to the same state add their rates. In $\mathbb{C}$ it is the interference condition, and it is the reason the quantum case is not a routine relabelling; see §??.

5 Propensity

Propensity is the real-weight execution of the decorated transition graph. Throughout this section and §?? we take $\mathcal{K} = \mathbb{R}_{\geq 0}$. The complex execution does not apply Definition ?? after replacing a weight by its modulus; it uses the amplitudes in the linear operator of §??.

The propensity of a rule at a refinement is where the classical construction is most easily got wrong, because four independent factors are involved and the literature usually presents some of them fused. We separate them.

Definition 21 (Propensity). Let $\mathbf{c} = (P, w)$ be a configuration, r a base rule and $\varphi \in \Phi_r$. Put

$$K(r, \varphi, P) = \{ (k, \sigma) \in \text{Red}_r(P) : \text{cls}_r(k, \sigma) = \varphi \}.$$

The *propensity* of (r, φ) at $\mathbf{c}$ is

$$\mathbf{a}(r, \varphi, \mathbf{c}) = w(\varphi) \cdot \sum_{(k, \sigma) \in K(r, \varphi, P)} g(k) \cdot \chi(r, k, \sigma),$$

where $\chi \in \{0, 1\}$ is the funding gate of §??. identically 1 in the absence of cost accounting. The *total propensity* is $\mathbf{a}_0(\mathbf{c}) = \sum_r \sum_{\varphi \in \Phi_r} \mathbf{a}(r, \varphi, \mathbf{c})$.

The four factors:

$w(\varphi)$ — **the rate constant**.

What kind of interaction this is, priced. Dynamic: this is the component update functions rewrite.

$g(k)$ — **the geometric factor**.

Where the interaction is, priced (Definition ??). Default 1.

The cardinality of the sum — multiplicity.

How many ways this interaction is available. This is Gillespie’s h_j , the combinatorial count of distinct reactant tuples [?], and it is the factor most often omitted in implementations, with the consequence that a term with fifty available redexes of a kind fires them no faster than a term with one.

χ — **the funding gate**.

Whether the interaction can be afforded (§??).

Proposition 22 (Multiplicity is well defined and computable). *For finite P , $|K(r, \varphi, P)|$ is finite, invariant under $\equiv$, and computable by enumerating matches of L in P modulo $\equiv$ and evaluating cls_r at each.*

Proof. Finiteness and $\equiv$ -invariance are Definition ?? together with Proposition ??; computability of the classification is ??. In the rho presentation the multiset structure of parallel composition is carried explicitly — MeTTaIL represents **PPar** as a hash bag [?] — so “matches modulo associativity and commutativity” is a bag operation rather than a search over permutations. $\square$

Proposition 23 (Total propensity is a sum over funded redexes). *With admissible key sets,*

$$\mathbf{a}_0(P, w) = \sum_r \sum_{(k, \sigma) \in \text{Red}_r(P)} w(\text{cls}_r(k, \sigma)) \cdot g(k) \cdot \chi(r, k, \sigma).$$

That is: every enabled redex contributes exactly once, at the rate of its own class.

Proof. The sets $K(r, \varphi, P)$ for $\varphi \in \Phi_r$ partition $\text{Red}_r(P)$, by Proposition ?? . Exchange the order of summation. $\square$

Proposition ?? is the payoff of the partition discipline in its most practical form. It says that the propensity computation may be organised *by redex* rather than *by key* — walk the term, find each redex, classify it, add its rate — with no risk of a redex being counted twice or missed. Without exclusivity the first walk over-counts; without exhaustiveness it under-counts. Neither error is visible in a trace; both silently distort the sampled distribution.

Remark 24 (Symmetry, and distinct redexes with a common contractum). Definition ?? counts selections from a multiset, which gives the symmetry factors of chemical kinetics for free: the $\binom{n}{2}$ of a dimerisation of like molecules is the number of unordered pairs drawn from a bag of n , and the $\text{In} \cdot \text{Out}$ of a binary interaction is the number of ways of drawing one reactant from each side. It does *not* handle the correction SPiM applies for an input and an output competing within a single summation, because the *rho* calculus has no summation operator; see Remark ??.

Distinct redexes may nonetheless have $\equiv$ -equal contracta — m indistinguishable pending messages on a channel give m redexes and one target. Classically the m contributions add as rates, so the rate to the target is m times the base rate, which is what Proposition ?? says. In the complex execution they add as amplitudes in the same matrix element of T_0 (§??). Equal phases therefore give constructive interference, while different refinement phases can partially or completely cancel. No normalisation is imposed to force the complex execution back to the classical multiplicity law.

6 Real weights: the Gillespie construction

6.1 The simulator

Definition 25 (Weighted GSLT simulator, $\mathcal{K} = \mathbb{R}_{\geq 0}$). Given an initial configuration $\mathbf{c}_0 = (P_0, w_0)$ and $t := 0$, repeat:

1. Compute $\mathbf{a}(r, \varphi, \mathbf{c})$ for all (r, φ) and $\mathbf{a}_0(\mathbf{c})$. If $\mathbf{a}_0 = 0$, halt.
2. Draw $u_1, u_2 \sim U(0, 1)$. Set $\tau := \mathbf{a}_0^{-1} \ln(1/u_1)$.
3. Select (r, φ) with probability $\mathbf{a}(r, \varphi, \mathbf{c})/\mathbf{a}_0(\mathbf{c})$, then select a position $(k, \sigma) \in K(r, \varphi, P)$ with probability $g(k)\chi(k)/\sum_{k'} g(k')\chi(k')$, using u_2 .
4. Fire: $P := k[R\sigma]$, and $w :=$ the fold of Definition ??, which in the default case is $u_i(w)$.
5. $t := t + \tau$.

Steps 1–3 and 5 are Gillespie’s direct method [?] verbatim. Step 4 is the addition, and the only one.

Theorem 26 (Markovianity on configurations). *The process of Definition ?? is a time-homogeneous continuous-time Markov chain on $\mathcal{C}$, with generator*

$$Q(\mathbf{c}, \mathbf{c}') = \gamma(\mathbf{c})(\mathbf{c}') \quad (\mathbf{c}' \neq \mathbf{c}), \quad Q(\mathbf{c}, \mathbf{c}) = -\sum_{\mathbf{c}' \neq \mathbf{c}} Q(\mathbf{c}, \mathbf{c}'),$$

for γ the coalgebra of Definition ?? . It is not in general a Markov chain on terms alone.

Proof. The propensity of Definition ?? is a function of $\mathbf{c} = (P, w)$ and of nothing else — in particular not of t or of the history — and the successor configuration is determined by $\mathbf{c}$ together with the selected redex. Waiting times are exponential with parameter $\mathbf{a}_0(\mathbf{c})$, which depends only on $\mathbf{c}$. This is the definition of a time-homogeneous CTMC with the stated generator. For the negative claim, it suffices to exhibit a theory and two runs reaching the same term with different maps, which we do at Example ??. $\square$

Remark 27 (Self-transitions, and why the diagonal is not $-\mathbf{a}_0$). Nothing in Definition ?? forbids a rule whose firing returns a configuration $\equiv$ -equal to the one it fired in. Such a redex contributes to $\mathbf{a}_0(\mathbf{c})$ but to no off-diagonal entry, so writing $Q(\mathbf{c}, \mathbf{c}) = -\mathbf{a}_0(\mathbf{c})$ — as the first version of this note did — gives a generator whose rows fail to sum to zero whenever a self-transition is enabled. The diagonal above is the correct one, and it is what an implementation should assemble.

The simulator of Definition ?? need not agree, and ours does not. It leaves $\mathbf{a}_0$ as computed and fires self-transitions like any others. A self-loop of rate λ is a fictitious jump in the sense of uniformisation, so the two conventions induce the same distribution of the state at every time and differ only in the number of recorded events, hence in sojourn statistics. Neither is wrong; mixing them is. An implementation should say which it reports, and §?? records that ours assembles Q by the formula above and samples with self-transitions retained.

Theorem ?? is the justification for Definition ??, and it is worth being explicit about the shape of the argument, because the design could have gone otherwise. If weight maps could change but were *not* part of the state, the process on terms would be non-Markovian: the rate of a transition out of P would depend on how P was reached. One would then be in the world of semi-Markov or generalised semi-Markov processes, with no exact simulation algorithm and no CTMC to model check. Putting the map in the state restores exactness at the price of a larger state space. That price is real — $\mathcal{C}$ is $\mathcal{T} \times \mathcal{K}^N$, a continuum — and §?? says when it is affordable. The reader should note that §?? runs this argument a second time, verbatim, one level up: the first version of this note took the lesson here and then declined to apply it in the quantum case, which is the single largest correction in this revision.

Example 28 (Dynamic weights are not a term-level phenomenon). Take the ρ theory with a single communication rule and keys $\Phi = \{\varphi_a, \varphi_b, \varphi_\perp\}$, where φ_x says the redex is a communication on the channel x . Let the update functions act on $w(\varphi_a)$ alone, by

$$u_a : w(\varphi_a) \mapsto w(\varphi_a) + 1, \quad u_b : w(\varphi_a) \mapsto 2 w(\varphi_a),$$

which do not commute. Put $Q = \text{for}(z \leftarrow a)\{\mathbf{0}\} \mid a!(\mathbf{0})$ and

$$P_0 = \text{for}(y \leftarrow a)\{Q\} \mid a!(\mathbf{0}) \mid \text{for}(y \leftarrow b)\{\mathbf{0}\} \mid b!(\mathbf{0}).$$

P_0 has exactly two redexes and both firing orders consume both, reaching the same term Q . But starting from $w_0(\varphi_a) = 1$, firing a first reaches Q with $w(\varphi_a) = (1+1) \cdot 2 = 4$, and firing b first reaches it with $w(\varphi_a) = 1 \cdot 2 + 1 = 3$. Q carries a single a -redex, so the two runs sit at the same term with expected waiting times $1/4$ and $1/3$. No rate function of the term alone reproduces both.

6.2 Summation-free SPiM is the static, name-keyed, ungeometric restriction

Theorem 29 (Conservativity over summation-free SPiM). *Instantiate the construction with*

(a) *the GSLT of the summation-free π -calculus, presented in MeTTaIL;*

- (b) key set $\Phi_{\text{COMM}} = \{\varphi_n : n \in \mathcal{N}\} \cup \{\varphi_\perp\}$, where φ_n holds of a redex just when it is a communication on the channel n ;
- (c) $w(\varphi_n) = \rho(n)$ the SPiM channel rate, $w(\varphi_\perp) = 0$;
- (d) all update functions the identity, all $g_c = 1$, all fold functions the second projection, and $\chi \equiv 1$.

Then Φ_{COMM} is admissible, the map is constant along every run, and for every process P the apparent rate of channel n computed by Definition ?? coincides with SPiM's:

$$\mathbf{a}(\text{COMM}, \varphi_n, P) = \rho(n) \cdot \text{In}_n(P) \cdot \text{Out}_n(P),$$

with $\text{In}_n, \text{Out}_n$ the counts of unguarded inputs and outputs on n . Hence the two simulators induce the same CTMC and the same trace distribution.

Proof. Admissibility. Distinct channels give disjoint sets of communication redexes, so (P1) holds; $\varphi_\perp$ completes (P2) by Remark ?. Each φ_n is expressible as $\text{in}(n)\top \mid \text{out}(n)\top$ at depth 2 in the structural fragment, so ?, ? and ? hold with $\alpha = 2$.

Staticity. All $u_i = \text{id}$ and all folds are second projections, so Definition ? gives $w' = w$.

The rate. With $g \equiv 1$ and $\chi \equiv 1$, Definition ? reduces to $\rho(n) \cdot |K(\text{COMM}, \varphi_n, P)|$. A communication redex on n is a choice of one unguarded input on n and one unguarded output on n in the parallel decomposition of P , taken modulo $\equiv$; the count is therefore $\text{In}_n(P) \cdot \text{Out}_n(P)$, which is SPiM's activity for n in the summation-free case [?].

Coincidence of the chains. Both simulators are Gillespie direct-method samplers over the same propensity function on the same state space (terms, the map being constant), so their generators agree and their trace distributions agree. $\square$

Remark 30 (What full SPiM would need). The restriction at (a) is not cosmetic and the theorem is now titled for it; the first version of this note advertised SPiM recovered “exactly” and relegated the fragment to a clause in the proof. SPiM's apparent rate in general subtracts $\sum_s \text{In}_n^s(P) \cdot \text{Out}_n^s(P)$ over summations s , excluding an input and an output that compete within the same choice. The ρ and summation-free π presentations have no choice operator, so that correction is identically zero and the equation above is the whole of the apparent rate.

Recovering full SPiM is a change to Definition ? rather than to the weighting. Presenting guarded choice as a term former gives $L^\sharp$ a summation shape, after which the selections from the multiset that count as redexes must exclude same-summation pairs, and the correction reappears exactly there — as a constraint on the redex set, not as a second rate. We have not carried this out, and until it is carried out the fragment should stay in the title.

Corollary 31 (Strictness). *The construction is strictly more expressive than SPiM in two independent directions. Relaxing (b) admits keys not of the form φ_n — for instance $\text{in}(n)\top \mid \text{out}(n)\text{@}\varphi$, “a communication on n whose payload names a process satisfying φ ,” which prices a transition by what is transferred and has no SPiM counterpart. Relaxing (d) admits $u_i \neq \text{id}$, and Example ? exhibits a term-level behaviour no static-rate calculus reproduces.*

We stress that the first direction is not merely a convenience. In a reflective calculus a name is a quoted process, so a key may inspect the name — $n \models \text{@}\varphi$ — and hence describe a *namespace* rather than a channel [?]. “Communications on any channel in the namespace of synapses of neuron i ” is a single key of bounded depth, where the name-keyed presentation would require one key per channel and would have to be regenerated when the network grew. §?? uses exactly this, and it is the honest form of the claim that a formula-valued key can see more than a channel name: what it sees more of is the name's own structure, not the term surrounding the redex.

6.3 Finiteness

Nothing above requires $\mathcal{C}$ to be finite; simulation only requires each step to be computable, and each step is. Model checking is another matter, and so is the quantum construction of §??, which builds operators on a space spanned by reachable configurations. We therefore record the condition, in two strengths that turn out not to be interchangeable:

Definition 32 (Bounded configuration space). A weighted GSLT with initial configuration $\mathbf{c}_0$ is *term-finite* when the set of terms reachable from P_0 is finite modulo $\equiv$, and *finite* when in addition the set of weight maps reachable from w_0 is finite.

Term-finiteness fails for a Turing-complete host in general and holds for the class of interest in §??, where it is a consequence of boundedness of a marked graph (Lemma ??). Map-finiteness fails as soon as an update function is a nontrivial contraction or dilation, as potentiation is; the standard remedies are to quantise the weights, to saturate them at a ceiling, or to model check the term-marginal only. We adopt saturation in §??, which is in any case what a bounded synapse does.

The distinction matters more than the first version of this note allowed. Classical simulation needs neither half. Classical model checking needs the second only if the property mentions weights. §?? uses both to obtain a finite matrix and computable defect operators, for the reasons given at Remark ??; what is a caveat here becomes a standing hypothesis there.

7 Complex weights: a postselected quantum transition operator

We now take $\mathcal{K} = \mathbb{C}$ and use the complex coalgebra coefficients as amplitudes. The classical and complex constructions therefore share the classified transition graph, but they do not share a propensity function. Definition ?? belongs to the real-weight instance; the complex instance starts from the weighted transitions of Definition ?? and adds their amplitudes linearly.

The construction has four parts. We first choose the Hilbert space of configurations (§??); then assemble the complex transition operator and normalise it to a contraction (§??); next we form its unitary defect dilation on one auxiliary qubit (§??); and finally we iterate the dilation with fresh auxiliary qubits, postselecting the all-zero branch to obtain a power of the transition operator (§??).

7.1 The space is a space of configurations

Definition 33 (Configuration Hilbert space). Let $\mathbf{c}_0 = (P_0, w_0)$ be finite in the sense of Definition ??, and let $\mathcal{B}$ be the set of configurations reachable from $\mathbf{c}_0$, with the term component taken modulo the full structural congruence $\equiv$. The *configuration Hilbert space* is

$$\mathcal{H} = \ell^2(\mathcal{B}), \quad \{ |\mathbf{c}\rangle : \mathbf{c} \in \mathcal{B} \}$$

with the displayed set as an orthonormal basis. A pure state is a unit vector $|\psi\rangle \in \mathcal{H}$; mixed inputs may equivalently be represented by density operators on $\mathcal{H}$.

The quotient by $\equiv$ is semantic, not a choice of dynamics. If $P \equiv Q$, then (P, w) and (Q, w) label the same basis vector. In particular, no Hamiltonian or other operator is introduced to move amplitude between representatives of one structural equivalence class: those representatives have already been identified. The only matrix entries below come from directed weighted rewrites between configuration classes.

The weight map remains part of the basis. If a firing updates w to w' , the target of the corresponding matrix entry is $|P', w'\rangle$, not the same term vector acted on by a newly built operator. Thus one operator contains the whole reachable plastic dynamics, just as one classical transition graph contains all reachable real-valued maps.

Remark 34 (Finiteness and boundedness). Finite reachability makes every operator below a finite matrix and makes its norm and defect operators unambiguous. The linear-algebraic construction extends to an infinite configuration space whenever the assembled transition operator is bounded, but that generalisation is not needed here. For the worked example, the capacity bound of §?? makes the term component finite and a finite amplitude alphabet makes the map component finite.

7.2 The complex transition operator and interference

For configurations $\mathfrak{c}, \mathfrak{c}' \in \mathcal{B}$, write $\text{Der}(\mathfrak{c}, \mathfrak{c}')$ for the set of triples (r, k, σ) whose firing, including the update and folds of Definition ??, carries $\mathfrak{c}$ to $\mathfrak{c}'$. The weight used by such a derivation is read from its *source* configuration.

Definition 35 (Complex transition operator). Define the unnormalised amplitude operator T_0 on basis vectors by

$$T_0 = \sum_{\mathfrak{c}, \mathfrak{c}' \in \mathcal{B}} \left(\sum_{(r, k, \sigma) \in \text{Der}(\mathfrak{c}, \mathfrak{c}')} w_{\mathfrak{c}}(\text{cls}_r(k, \sigma)) g(k) \chi(r, k, \sigma) \right) |\mathfrak{c}'\rangle \langle \mathfrak{c}|,$$

where $w_{\mathfrak{c}}$ is the map component of $\mathfrak{c}$ and $\chi \equiv 1$ when cost accounting is absent. Put

$$\alpha = \max\{1, \|T_0\|\}, \quad T = \alpha^{-1} T_0.$$

Then T is a contraction: $T^\dagger T \leq I$.

The normalisation is global. A bound such as $|w(\varphi)| \leq 1$ on each entry would not imply $\|T_0\| \leq 1$, because several derivations can add in one matrix element and several targets can leave one source. Conversely, dividing by α changes neither relative phases nor the normalised postselected output: whenever $T_0|\psi\rangle \neq 0$,

$$\frac{T|\psi\rangle}{\|T|\psi\rangle\|} = \frac{T_0|\psi\rangle}{\|T_0|\psi\rangle\|}.$$

It changes only the probability with which that output is obtained. An implementation may replace $\|T_0\|$ by any efficiently computable upper bound; the resulting contraction has the same conditional state and a correspondingly smaller success probability.

Proposition 36 (Complex weights act before measurement). *For every source and target configuration,*

$$\langle \mathfrak{c}' | T | \mathfrak{c} \rangle = \alpha^{-1} \sum_{(r, k, \sigma) \in \text{Der}(\mathfrak{c}, \mathfrak{c}')} w_{\mathfrak{c}}(\text{cls}_r(k, \sigma)) g(k) \chi(r, k, \sigma).$$

Thus derivations with a common target add as complex amplitudes. Relative phases may reinforce or cancel before any absolute square is taken.

This is where the complex codomain does work. Two contributions z and $-z$ to one matrix entry cancel; contributions z and z add constructively. Refinement classes remain useful because they supply the amplitudes, but they do not partition the target matrix entries: derivations from different rules or different classes can reach the same configuration class and therefore interfere there. The complex construction is not required to reproduce the real propensity as a population marginal; it is a different execution of the same classified rewrite graph.

7.3 Defects and the one-step unitary dilation

A non-unitary T cannot by itself be the evolution of a closed quantum system. Its failure to be an isometry is measured by a defect operator.

Definition 37 (Defect operators). For the contraction T , define

$$D_T = (I - T^\dagger T)^{1/2}, \quad D_{T^\dagger} = (I - TT^\dagger)^{1/2}.$$

Both are positive operators. The first measures norm lost by applying T to an input state; the second completes the action on the other auxiliary-qubit sector.

Theorem 38 (Unitary defect dilation). On $\mathbb{C}^2 \otimes \mathcal{H} \cong \mathcal{H} \oplus \mathcal{H}$, the block operator

$$U_T = \begin{pmatrix} T & D_{T^\dagger} \\ D_T & -T^\dagger \end{pmatrix}$$

is unitary.

Proof. The singular-value decomposition of T gives the intertwining identities $TD_T = D_{T^\dagger}T$ and $T^\dagger D_{T^\dagger} = D_T T^\dagger$. Block multiplication then gives

$$U_T^\dagger U_T = \begin{pmatrix} T^\dagger T + D_T^2 & T^\dagger D_{T^\dagger} - D_T T^\dagger \\ D_{T^\dagger} T - TD_T & D_{T^\dagger}^2 + TT^\dagger \end{pmatrix} = I,$$

and the same identities give $U_T U_T^\dagger = I$. $\square$

In the finite case of Definition ??, the same singular-value decomposition used to obtain $\|T_0\|$ computes the dilation explicitly. If $T = V\Sigma W^\dagger$, then

$$D_T = W(I - \Sigma^2)^{1/2}W^\dagger, \quad D_{T^\dagger} = V(I - \Sigma^2)^{1/2}V^\dagger.$$

Thus the construction requires one matrix decomposition, two diagonal square roots and the assembly of the four blocks of U_T .

Remark 39 (Why there are two canonical defects). The often-used shorthand

$$\begin{pmatrix} T & D_T \\ D_T^\dagger & -T^\dagger \end{pmatrix}$$

is exact with one positive defect when T is normal, since then $D_T = D_{T^\dagger} = D_T^\dagger$. A transition operator assembled from a directed rewrite graph need not be normal. For a generic T , putting the same positive square root in both off-diagonal blocks does not give a unitary; the two-defect formula of Theorem ?? is the canonical form. It has precisely the same success and failure branches on an auxiliary input $|0\rangle$.

Definition 40 (One postselected quantum step). Append an auxiliary qubit in $|0\rangle$ and apply U_T . For every $|\psi\rangle \in \mathcal{H}$,

$$U_T|0, \psi\rangle = |0, T\psi\rangle + |1, D_T\psi\rangle.$$

The outcome 0 is success and leaves the unnormalised system state $T|\psi\rangle$; the outcome 1 is failure and leaves the failed state $|\phi\rangle = D_T|\psi\rangle$.

For a normalised input,

$$p_{\text{succ}}(\psi) = \|T\psi\|^2, \quad p_{\text{fail}}(\psi) = \|D_T\psi\|^2 = 1 - \|T\psi\|^2.$$

If T is unitary, both defects vanish and success is certain. More generally, the failure probability is exactly the expectation of $I - T^\dagger T$, so it is small on states for which T is close to an isometry.

7.4 Iteration and all-zero postselection

One auxiliary qubit records whether one application of T succeeded. A coherent implementation of n applications uses n fresh auxiliary qubits, so that a failed branch cannot later return to the all-success record.

Definition 41 (Iterated dilation). Let $U_T^{(j)}$ act as U_T on the system and auxiliary qubit j , and as the identity on the other $n - 1$ auxiliaries. Define

$$W_n = U_T^{(n)} U_T^{(n-1)} \dots U_T^{(1)}.$$

All auxiliary qubits are initialised in $|0^n\rangle$.

Theorem 42 (The all-zero branch is T^n). *For every $|\psi\rangle \in \mathcal{H}$,*

$$W_n|0^n, \psi\rangle = |0^n, T^n\psi\rangle + |\text{fail}_n(\psi)\rangle,$$

where

$$(|0^n\rangle\langle 0^n| \otimes I)|\text{fail}_n(\psi)\rangle = 0.$$

Equivalently,

$$(\langle 0^n| \otimes I)W_n(|0^n\rangle \otimes I) = T^n.$$

Hence postselecting the all-zero auxiliary outcome yields the unnormalised state $|0^n, T^n\psi\rangle$ with probability

$$p_n(\psi) = \|T^n\psi\|^2,$$

and, when $p_n(\psi) > 0$, the conditional system state is $T^n|\psi\rangle/\|T^n|\psi\rangle\|$.

Proof. For $n = 1$ this is Definition ???. Suppose the all-zero component after j steps is $|0^j, T^j\psi\rangle$. The next unitary acts on a fresh qubit and maps that component to $|0^{j+1}, T^{j+1}\psi\rangle$ plus a term whose new qubit is 1. Every branch that had already failed retains a 1 in an earlier qubit because later unitaries do not act on it. Induction proves both statements. $\square$

Since $T = \alpha^{-1}T_0$, the all-success component is $\alpha^{-n}T_0^n|\psi\rangle$. The normalised conditional state is therefore the one defined by the original complex weights, while the normalisation contributes the known success factor α^{-2n} . For a mixed input ρ , the all-zero branch is the trace-nonincreasing map

$$\rho \mapsto T^n \rho (T^\dagger)^n,$$

with success probability $\text{Tr}(T^n \rho (T^\dagger)^n)$.

7.5 The generated logic supplies canonical measurements

The generated spatial-behavioural logic still gives a canonical family of measurements, but it is now applied to the postselected discrete-step state rather than to a separately postulated quantum continuous-time Markov chain.

Proposition 43 (Basis-diagonal projectors). *For each formula φ of the generated logic, put*

$$\Pi_\varphi = \sum_{\substack{\mathbf{c} \in \mathcal{B} \\ \mathbf{c} \models \varphi}} |\mathbf{c}\rangle\langle \mathbf{c}|.$$

Then Π_φ is an orthogonal projector. For an admissible key set Φ_r , the projectors $\{\Pi_\varphi\}_{\varphi \in \Phi_r}$ are mutually orthogonal and sum to $\Pi_{L^\sharp}$; adjoining $\Pi_{\neg L^\sharp} = I - \Pi_{L^\sharp}$ gives a projective measurement of $\mathcal{H}$.

Proof. Each Π_φ is a sum of rank-one projectors onto distinct basis vectors. Exclusivity (P1) gives pairwise orthogonality, and exhaustiveness (P2) says that every basis vector satisfying $L^\sharp$ occurs in exactly one sum. The complementary projector completes the identity. $\square$

For a pure input, the probability of φ after n successful steps is

$$\Pr(\varphi \mid 0^n) = \frac{\langle \psi | (T^\dagger)^n \Pi_\varphi T^n | \psi \rangle}{\langle \psi | (T^\dagger)^n T^n | \psi \rangle},$$

provided the denominator is nonzero. For a mixed input, replace the numerator and denominator by the corresponding traces. The projectors are generated from the presentation; the dynamics is generated from the same presentation through T ; and no additional Hamiltonian is required to make the complex amplitudes observable.

8 Worked example: an asynchronous stochastic spiking network

8.1 The design decision that makes the example an example

One can always model a neural network in a process calculus by writing the arithmetic into the payloads: send a real number down a channel, multiply it by a stored weight, sum, squash, forward. Nothing in this note would be doing any work; the calculus would be a programming language and the network a program written in it.

We do the opposite. The synaptic efficacies are *not* in the terms. They are the values of the weight map, keyed by spatial-behavioural formulae picking out the synapses. In one sentence, the thesis of this section is that *synaptic efficacy is represented as transition intensity in the operational semantics, and the same transition updates that intensity*. The consequences are that firing is a stochastic event rather than an arithmetic one, that the network's response nonlinearity is a property of the chain rather than a function written down anywhere, and — the reason for choosing this example — that synaptic plasticity is literally the update functions of Definition ???. A network that learns and a network that infers are the same weighted GSLT running the same transition relation.

We are correspondingly careful about what the object is. It is an asynchronous stochastic spiking recurrent network with plastic transition rates. It is not a recurrent neural network in the machine-learning sense, and the first version of this note called it one, which invites a comparison it does not satisfy: there is no backpropagation, no membrane potential, no refractory period, no global clock and no discrete timestep, and the quantity compared with a textbook artificial neuron's pre-activation in §?? is a hazard rate rather than an affine form awaiting an activation function. Nothing in the argument needs the stronger name, and the claim it does support is the more interesting one.

8.2 Neurons are persistent for-comprehensions

Definition 44 (Neuron). Let neuron i have presynaptic channels $a_{1i}, \dots, a_{k_i i}$ (its dendrites), an axonal channel b_i , and postsynaptic targets $\text{post}(i)$. Its *threshold presentation at θ_i* is the persistent for-comprehension

$$N_i \triangleq \text{for} \left(\underbrace{x_{s_1} \Leftarrow a_{s_1 i} \ \& \ \dots \ \& \ x_{s_\theta} \Leftarrow a_{s_\theta i}}_{\text{one join group per } \theta\text{-subset } S=\{s_1, \dots, s_\theta\}} ; \dots \right) \{ \prod_{j \in \text{post}(i)} a_{ij}! (\text{spk}) \},$$

with one join group per θ_i -element subset of the dendrites, $\&$ joining within a group and ; separating groups. The network is $\text{Net} \triangleq \text{new } \vec{a} \text{ in } \{ \prod_i N_i \mid M_0 \}$, where M_0 is an initial multiset of pending spikes.

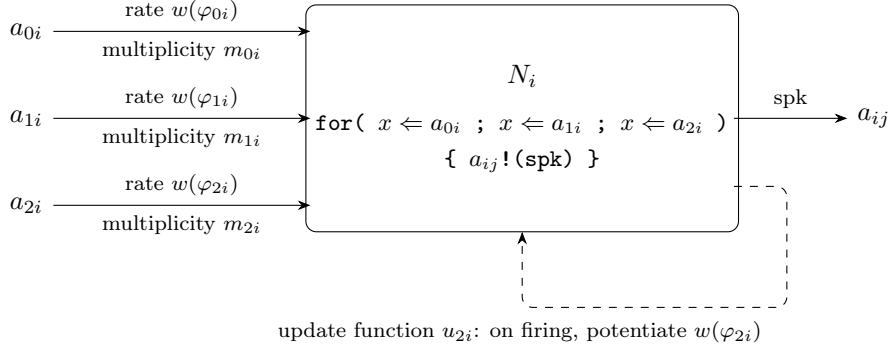


Figure 1: A neuron. The three factors of Definition ?? are visible: the weight map supplies a rate constant per synapse, the marking supplies multiplicity, and the geometric factor is 1. The dashed arrow is the refinement entry’s update function, which is where plasticity lives.

Three points about this definition, in increasing order of importance.

The receipts are *persistent* ($\Leftarrow$, not $\leftarrow$). A linear receipt evaporates after one firing, so a network built from linear receipts models a single sweep, not a network. The same distinction was load-bearing in the knot encoding of [?], and for the same reason: persistence is what makes the object a standing structure rather than a trace through one. Rholang receipts are linear-or-persistent as a whole, so the pending spikes are still consumed per firing while the neuron itself survives.

Recurrence is not an additional construct. It is the statement that b_i — equivalently, some a_{ij} for $j \in \text{post}(i)$ — feeds a dendrite of a neuron from which i is reachable, i itself included. The unrolled network in the machine-learning sense is the reduction graph; the hidden state at a given moment is the *marking*, the standing multiset M of pending spikes on the restricted channels. Since the marking is part of the term and the term is part of the configuration, hidden state, weights and time are carried by one object.

And the threshold is in the join structure, which is the next subsection.

8.3 The keys, and why they form a partition

Definition 45 (Synaptic keys). For each synapse (j, i) put

$$\varphi_{ji} \triangleq \text{in}(a_{ji})\top \mid \text{out}(a_{ji})\top,$$

the formula holding of a communication redex on a_{ji} , and let $\Phi_{\text{COMM}} = \{\varphi_{ji}\} \cup \{\varphi_{\perp}\}$ with $\varphi_{\perp} = L^{\#} \wedge \neg \bigvee_{ji} \varphi_{ji}$.

Proposition 46. Φ_{COMM} is admissible, with $\alpha = 2$.

Proof. Distinct synapses are distinct channels, and a communication redex is on exactly one channel, so (P1) holds; $\varphi_{\perp}$ gives (P2) by Remark ??. Each key has depth 2 by Definition ??, and checking $\text{in}(a)\top \mid \text{out}(a)\top$ against a term is a bag lookup, so ?? and ?? hold. $\square$

The weight $w(\varphi_{ji})$ is the efficacy of synapse (j, i) . Nothing else in the presentation mentions it.

8.4 Threshold is join structure

Proposition 47 (McCulloch–Pitts, as an expressivity lemma). *The threshold presentation of Definition ?? realises the monotone Boolean threshold function $\mathbf{1}[\sum_j \mathbf{1}[a_{ji} \text{ populated}] \geq \theta_i]$:*

the neuron has an enabled redex exactly when at least θ_i of its dendrites carry a pending spike. It uses $\binom{k_i}{\theta_i}$ join groups.

Proof. A group S is enabled iff every a_{si} for $s \in S$ carries a spike, since $\&$ requires simultaneous availability of all its binds; the receipt is enabled iff some group is, since $;$ separates alternatives. Some θ_i -subset is fully populated iff at least θ_i dendrites are populated. The count of groups is the count of θ_i -subsets. $\square$

So the historical origin of the neuron model — McCulloch and Pitts’ threshold unit [?] — is not encoded in a weighted GSLT but *is* a for-comprehension, with the threshold appearing as the arity of the joins and the alternatives as the group separator. We now state this as an expressivity lemma and not, as the first version implied, as a practical construction, and the qualification is not cosmetic. $\binom{k}{\theta}$ groups is unusable at realistic fan-in, so the scalable presentation is the one at $\theta = 1$, where threshold behaviour is trivial and the join structure carries no weight. What the lemma establishes is that no apparatus beyond the for-comprehension is needed to *express* a threshold; what it does not establish is that this is how one would build a large network.

Remark 48 (The **where** clause). Rholang 1.4’s for-comprehension carries a **where** clause [?], which collapses the presentation to a single group at the cost of changing the semantics. The guarded form binds all k dendrites in one group, waits for *all* of them, and only then applies the filter ϕ . That is a coincidence detector with a filter, not a threshold unit. The two presentations agree at $\theta = k$ and differ elsewhere, and which one a modeller wants is a modelling question. We use $\theta_i = 1$ below, where both are trivial and the group presentation has k_i single-bind alternatives.

Remark 49 (Coincidence versus rate). $\theta_i = k_i$ gives a pure coincidence detector: the neuron fires only on simultaneous arrival at every dendrite. $\theta_i = 1$ gives a pure rate integrator: any arrival can trigger it, and the weights determine how fast. Real cortical neurons sit between, and so does the presentation, with θ as the dial. This is a rare case in which a biological parameter and a syntactic parameter are the same parameter.

8.5 The weighted sum, and where the squashing function went

Take $\theta_i = 1$, so N_i has k_i single-bind alternatives, and let the marking assign m_{ji} pending spikes to dendrite a_{ji} .

Proposition 50 (Propensity of a neuron). *With $g \equiv 1$ and $\chi \equiv 1$, the total propensity of the redexes belonging to neuron i is*

$$\mathbf{a}_i = \sum_j w(\varphi_{ji}) \cdot m_{ji}.$$

Proof. The receipt is persistent, so exactly one input is available on each a_{ji} , and the marking supplies m_{ji} outputs; by Definition ?? the number of redexes classified φ_{ji} is $1 \cdot m_{ji}$. Definition ?? gives $w(\varphi_{ji})m_{ji}$ per synapse, and the classes are disjoint by Proposition ??, so Proposition ?? lets them be summed. $\square$

Formally this is the weighted sum $\sum_j w_{ji}m_{ji}$ of the textbook artificial neuron, with the presynaptic activity m_{ji} read as the marking, and it has not been written into any term; it is what Definition ?? computes. The comparison should nevertheless be made carefully, because the two quantities are not the same kind of thing. $\mathbf{a}_i$ is a hazard rate, with units of inverse time, governing a race between exponential clocks. It is not an affine pre-activation waiting to be passed through an activation function, and the first version of this note wrote as

though it were. What the next corollary shows is that a saturating nonlinearity nevertheless appears, and appears without being supplied.

Corollary 51 (The response saturates and nobody wrote it). *Hold the marking on i 's dendrites fixed over a window $[0, \Delta]$. The probability that neuron i fires at least once in the window is*

$$\Pr[i \text{ fires}] = 1 - \exp(-\Delta \sum_j w(\varphi_{ji})m_{ji}),$$

a monotone function of the weighted sum, with slope Δ at the origin and asymptote 1.

Proof. Under a marking held fixed on i 's dendrites the propensity $\mathbf{a}_i$ is constant, so firings of i within the window are the events of a homogeneous Poisson process of rate $\mathbf{a}_i$, by Theorem ?? restricted to the sub-chain along which $m_{\cdot i}$ does not change. The probability of at least one event in $[0, \Delta]$ is $1 - e^{-\mathbf{a}_i \Delta}$. $\square$

Two provisos, one standard and one easy to miss.

The standard one is that the marking is not in fact frozen, since other neurons are firing into it. Corollary ?? is a statement about a window short enough that the input has not moved, which is the usual justification for a rate-coded neuron and the usual approximation.

The one that is easy to miss — and that the first version's proof did miss, freezing only the other neurons — is that i 's own firings change the marking. A firing consumes a spike from the dendrite it fired on, so $\mathbf{a}_i$ drops and the process is not homogeneous past the first event unless the marking is externally replenished. The hypothesis must therefore be read as freezing $m_{\cdot i}$ against *all* sources of change, i 's own consumption included. That is exactly the standing-input regime in which a rate code is normally discussed, and it is why the statement is about the first event rather than about a count.

Remark 52 (Saturating, not logistic). $1 - e^{-\Delta x}$ is an exponential saturation function. It is not the logistic $\sigma(x) = (1 + e^{-x})^{-1}$, and the first version of this note said that it was — in this section, and in its abstract, where the emergence of a logistic response was offered as a selling point. The correct claim is worth as much and should be stated exactly: *a saturating nonlinear activation emerges from exponential race semantics without being programmed into the term*. No squashing function was supplied and none is needed. A neuron under enormous drive still cannot fire more than once per event, so its firing probability per window is bounded by 1 however large the weighted sum grows, and that bound is the waiting-time law of the chain rather than a modelling choice.

8.6 Boundedness

Lemma 53 (Boundedness). *Let $f_i = |\text{post}(i)|$ be the fan-out of neuron i . If $f_i \leq \theta_i$ for every i , the total number of pending spikes is non-increasing along every run, the reachable term set is finite, and **Net** is term-finite in the sense of Definition ??.*

Proof. Firing N_i consumes θ_i pending spikes (one per bind in the selected group) and produces f_i . The total marking therefore changes by $f_i - \theta_i \leq 0$. Since the neurons are persistent and the channel set is fixed, a term reachable from **Net** is determined by its marking, and markings with at most $|M_0|$ tokens over finitely many channels are finite in number. $\square$

The hypothesis is restrictive: it forbids fan-out exceeding threshold, which most networks want. Two standard repairs, both expressible in the theory:

1. *Capacity.* Guard the neuron body's sends by a **where** clause bounding the occupancy of the target channel *after* the firing by c . The marking is then bounded by c per channel and the state space by $c^{|\vec{a}|}$, restoring term-finiteness unconditionally. This is a saturating

synapse, and it is what a real one does. The post-firing reading is not fussiness: the obvious alternative, refusing a send when the target is already full, blocks the consumer and deadlocks the network (§??).

2. *Leak*. Add a base rule $a_{ji}!(\text{spk}) \rightsquigarrow \mathbf{0}$ with its own key and weight λ_{leak} . This does not restore term-finiteness by itself but makes the marking positive-recurrent, which is enough for stationary analysis and classical simulation, though it falls outside the finite-matrix treatment adopted in §??.

We take capacity in what follows, since §?? needs a finite-dimensional $\mathcal{H}$.

Remark 54 (The network is a marked graph). Under Lemma ??, **Net** is a live, bounded, conservative marked graph in the Petri-net sense, and the analysis of [?] applies verbatim: a *period* is a firing schedule returning the marking to its initial value, and it is the right analogue of a timestep. The weighted-GSLT layer adds what a marked graph does not have, namely a probability distribution over which of the enabled transitions fires next and how long it takes.

8.7 Plasticity is the update function

Definition 55 (Local potentiation entry). Equip the communication rule’s entry at φ_{ji} with

$$u_{ji}(w) = w[\varphi_{ji} \mapsto \min\{w_{\max}, w(\varphi_{ji}) + \eta\}],$$

for a potentiation increment $\eta > 0$ and ceiling $w_{\max}$. The ceiling is not cosmetic: with a quantised increment it is what makes the reachable weight-map set finite, which Remark ?? requires of §??.

So the augmented rule reads, in full:

```
COMM . |- (PPar {(PForUser rows p), (POutput n q), ...rest})
  ~> (PPar {(subst ...), ...rest}) [1.0] {
    phi(0,1) => (w_01, hebb(0,1)),
    phi(1,1) => (w_11, hebb(1,1)),
    ...
    phi_bot => (0.0, id)
  };
ParCong . | S ~> T |- (PPar {S, ...rest}) ~> (PPar {T, ...rest}) [1.0] { snd };
```

Theorem 56 (Learning and inference are one relation). *Let **Net** be a weighted GSLT as above. The transition relation of Definition ?? on $\mathcal{C} = \mathcal{T} \times \mathcal{W}$ has two marginals: the projection to $\mathcal{T}$ is the network’s inference dynamics, and the projection to $\mathcal{W}$ is its learning dynamics. Neither is a separate process. In particular there is no training schedule, no separate learning clock, and no distinguished learning phase; the learning rate η is not a hyperparameter of an outer loop but a component of the theory, sitting in the same refinement entry as the weight it modifies.*

Proof. Immediate from Definitions ?? and ??: a single step produces (P', w') with P' obtained by the rewrite and w' by the fold of the update function. $\square$

This is the claim the example exists to make, and it is exactly what a static-rate calculus cannot express. In SPiM a rate is fixed at declaration, so a stochastic π model of a neural network can model the firing but not the learning; one must step outside the calculus, adjust the rates, and re-run. Here the adjustment is a transition of the same system.

Remark 57 (How Hebbian this is). Definition ?? potentiates φ_{ji} when a communication on a_{ji} fires — that is, when a presynaptic spike is consumed by an enabled postsynaptic receipt. The event is local and activity-dependent, and the condition “pre and post were jointly involved” is not a correlation computed by an observer but the firing of a single rewrite, which by construction requires both parties. That much is the structural content of Hebb’s proposal [?] and it is the part worth having: the event which transmits is the event which potentiates, because a communication is one rewrite.

It is nevertheless less than Hebbian learning as usually meant, and the first version of this note was too generous in saying that cells which fire together wire together *literally*. Hebbian plasticity invokes a correlation between pre- and *postsynaptic firing*; successful transmission into an enabled receipt is not by itself postsynaptic firing, since the receiving neuron may not reach threshold. What Definition ?? gives is a minimal local Hebbian potentiation rule, and §?? says what the timing-dependent version costs.

8.8 Spike-timing dependence needs a wider update signature

Spike-timing-dependent plasticity makes Δw a function of the interval between pre- and postsynaptic spikes, potentiating for pre-before-post and depressing for the reverse [?]. The ingredients are available — the simulator of Definition ?? produces a timestamp at every step — but the update function of Definition ?? cannot see them, since its type is $\mathcal{W} \rightarrow \mathcal{W}$.

Definition 58 (Extended update). Replace $u_i : \mathcal{W} \rightarrow \mathcal{W}$ by

$$u_i : \mathcal{W} \times \text{Match} \times \partial\mathcal{T} \times \mathbb{R}_{\geq 0} \times \mathcal{S} \longrightarrow \mathcal{W},$$

taking additionally the matched substitution σ , the position k at which the rule fired, the current simulation time t , and a bounded *trace summary* $\mathcal{S}$ — for STDP, one eligibility trace per synapse, itself decaying at a rate specified in the theory.

Theorem ?? survives this extension provided $\mathcal{S}$ is finite-dimensional and updated deterministically, since one simply enlarges the configuration to $(P, w, \mathcal{S})$; it does *not* survive an update function reading unbounded history, which would make the process genuinely non-Markovian and put exact simulation out of reach. The eligibility-trace formulation of STDP is precisely the standard device for staying inside the bounded case, and it is pleasant that the constraint arrives here from the requirement that the chain be a chain rather than from computational convenience.

8.9 Uniformity: one theory for every network

Definition ?? gives one key per synapse, so a network with 10^4 synapses has 10^4 keys and a network that grows needs a new theory. Reflection removes both problems. Following the construction of [?], take the synapse channels to be structured names

$$a_{ji} = @[*net, j, i],$$

so that a name predicate can recover the endpoints. Then

$$\varphi_{\text{syn}} \triangleq \exists j, i. \text{in}(@[*net, j, i])\top \mid \text{out}(@[*net, j, i])\top$$

is a single key describing the whole *namespace* of synapses of this network, which is namespace logic doing what it was built for [?]. The weight map then has one entry per network rather than per synapse; per-synapse efficacies are recovered by letting the entry’s value be a function of the recovered indices, which is the standard move of trading a large map for a small map

with structured keys. A greatest fixed point makes “every neuron reachable from i eventually fires” a single formula rather than an n -fold conjunction, which is the uniformity ν buys [?, §3]. In an atomic-name calculus none of this is available: the indices would have to travel as message payloads and the logic could reach them only by observing traffic.

8.10 The postselected quantum network

Replace the real efficacies by amplitudes $z_{ji} \in \mathbb{C}$. These amplitudes are not converted to $|z_{ji}|^2$ rates. Each enabled synaptic communication contributes its complex value directly to the corresponding entry of T_0 in Definition ??; derivations with the same successor configuration add before any measurement. The capacity bound is retained, and the complex update functions are required to range over a finite amplitude alphabet $\mathcal{Z} \subset \mathbb{C}$. The real potentiation update of Definition ?? is therefore replaced, in this instance, by a chosen map $u_{ji} : \mathcal{Z}^N \rightarrow \mathcal{Z}^N$ that may change both modulus and phase.

The example then satisfies Definition ??: capacity makes the reachable term set finite, and the finite alphabet makes the reachable map set finite. The complete reachable graph is assembled once into T_0 , normalised to the contraction T , and dilated by Theorem ?. One synaptic step uses one auxiliary qubit; n coherent steps use n fresh auxiliaries; and postselecting their all-zero outcome yields T^n by Theorem ?.

Remark 59 (Multiplicity becomes interference). By Remark ?, m indistinguishable pending spikes on a_{ji} can give m derivations with a common contractum. In the complex network their contributions occupy one matrix element. If they carry the same amplitude and geometric factor, that element is multiplied by m and its contribution to a measured probability is proportional to m^2 ; if different refinement classes contribute opposite phases, it can instead be suppressed or cancelled. This is not a claimed classical rate law. It is the direct consequence of treating the weights as amplitudes before measurement, and it is exactly the behaviour the former magnitude-square construction removed.

Structural congruence does not supply an additional coherent channel. Configurations whose terms are $\equiv$ -equal are one basis state, so there is no Hamiltonian mixing alternative representatives. All coherence in this example is already visible in the sums defining T_0 .

The projectors of Proposition ? remain available. For example, $\Pi_{\varphi_{\text{syn}}}$ measures whether a postselected configuration has an available synaptic transmission. After n successful steps its expectation is given by the conditional formula of §??.

8.11 What this example does not show

Honesty about scope. The learning here is local and Hebbian; nothing above implements backpropagation, which needs a global error signal propagated against the direction of inference. One can encode such a signal as traffic on a second family of channels with its own keys, and the apparatus does not forbid it, but the result would no longer have the property that makes Theorem ? interesting — the error channel is an outer loop wearing a costume. Whether a genuinely local weighted-GSLT learning rule can match backpropagation is not a question this note answers, and the honest statement is that the example demonstrates a *mechanism* for plasticity inside the calculus, not a competitive learning algorithm.

Nor do we claim biological fidelity. The neuron of Definition ? has no membrane potential, no refractory period, and no dendritic geometry. The first two are easy additions — a potential is a payload, a refractory period is a guard — and the third is what the geometric factor g of §?? is for, which is a use we have not developed.

9 Implementation

9.1 Surface syntax

The weighting is a decoration of MeTTaIL’s `rewrites` block, and it should read as one. The existing surface is

```
rewrites {
  Exec . |- (PDrop (NQuote P)) ~> P;
  ParCong . | S ~> T |- (PPar {S, ...rest}) ~> (PPar {T, ...rest});
}
```

and the weighted surface adds a bracketed weight and a brace block after the rule, in both cases optional:

```
rewrites {
  // base rule: [scalar] { key => (weight, update) }
  COMM . |- (PPar {(PForUser rows p), (POutput n q), ...rest})
    ~> (PPar {(subst rows p q), ...rest}) [1.0] {
    in(n)TOP | out(n)@fast => (2.5, scale(0.9)),
    in(n)TOP | out(n)TOP => (0.7, id),
    default => (0.0, id)
  };

  // base rule, unweighted: defaults to [1.0] { default => (1.0, id) }
  Exec . |- (PDrop (NQuote P)) ~> P;

  // context rule: [geometric factor] { fold }
  ParCong . | S ~> T |- (PPar {S, ...rest}) ~> (PPar {T, ...rest}) [1.0] { snd
  };
}
```

Notes on the design. `default` is sugar for the completion of Remark ?? — the elaborator synthesises $L^\sharp \wedge \neg \bigvee_i \varphi_i$ — so a modeller writes exclusive keys and gets a partition. Exclusivity itself is checked at elaboration time and is a hard error, not a warning, since Proposition ?? fails without it. Keys are written in the concrete syntax of the generated logic; the two shown are ordered from more to less specific purely for readability, since exclusivity means order is semantically irrelevant. A context rule’s default factor is 1 and its default fold is `snd`, which together implement “addressing is free and does not touch the map” (§??).

9.2 What is built

The first version of this note measured itself against a prototype called `mettail-gillespie`, whose `TermRef` was an opaque identifier, whose spatial behaviours were a hand-rolled enum, whose rates were clamped to the unit interval, and whose quantum module was amplitude bookkeeping over a classical branching structure. That crate no longer exists. It has been superseded by `rho-weighted` [?]. The classical simulator and exhaustive graph builder are the implementation basis used below. The unitary-dilation construction of §?? is a new specification target: the old amplitude bookkeeping and the former density-operator path do not constitute an implementation of it.

What the crate is. A *simulator*, and deliberately not the interpreter. The interpreter maintains the state of a running shard; the simulator answers what would happen if, across a family of hypothetical configurations, and yields transition graphs, generators and parameter

sweeps rather than a single history. That separation is what makes exhaustive graph construction, greatest fixed points and the finite matrix construction required by §?? affordable here and unaffordable there, and it is why the crate owns its own space — an immutable channel index with cheap forking — rather than a tuple space with checkpoints. It shares the interpreter’s matching semantics without sharing its code paths, which makes agreement between them an obligation rather than a theorem; see below.

Correspondence with this note.

Specification	Implementation
Definitions ??–??, admissibility	partition and fragment checks; failure is an error, never a fallback
Proposition ??, classification	the classifier, total by construction
Definition ??, propensity	organised by redex, per Proposition ??
Definition ??, the SSA	the sampler, with provenance recorded per run
Theorem ??, the generator	exhaustive graph plus Q , diagonal per Remark ??
Definition ??, updates	the update context: bindings, position, clock, trace
§??, the network	the worked example, end to end
Definitions ??–??, the quantum path	outstanding: sparse T_0 , global normalisation, defects, dilation

Three constraints an implementation exposed. These are the reason this section is worth reading rather than a table of ticks.

1. *Multiplicity is invisible in a trace.* The superseded prototype omitted the cardinality of $K(r, \varphi, P)$ entirely, and its traces looked entirely plausible; nothing an eye could catch distinguishes a chain running at the wrong rate from one running at the right rate. Acceptance therefore has to be statistical rather than observational. The birth–death theory is checked against a truncated Poisson law, with a control asserting that the Poisson and geometric laws are half a unit of total variation apart, so that the test cannot pass vacuously against a chain that ignores multiplicity.
2. *A capacity guard must be evaluated post-firing.* The obvious reading of the guard of §?? — refuse a send when the target channel is already at capacity — blocks the consumer and deadlocks the network. The guard must bound occupancy *after* the firing it gates. Relatedly, a hard capacity makes a token amplifier absorbing rather than live, which is a stop and not a deadlock; a network that is to saturate and keep running wants the leak rule of §?? instead, trading term-finiteness for positive recurrence and thereby giving up §??.
3. *Contractivity is global.* Entrywise bounds on complex weights do not bound the norm of the assembled transition operator, because amplitudes can add within a column and within a target entry. A quantum backend must build or bound $\|T_0\|$ first, then construct T , D_T , and $D_{T^\dagger}$; dilating each weight independently implements a different map.

What is outstanding. Four items follow.

1. *The logic is hand-written, not generated.* The connectives of §?? are implemented by hand behind a port whose interface is the one an OSLF-generated instance would implement, so the swap is at the boundary rather than through the simulator, which was the design intent. Until it lands, the genericity over GSLTs claimed in §?? is argued and not demonstrated, and that is the largest gap in the note.

2. *Surface syntax.* §?? is a design and not a grammar. It needs parser and tree-sitter changes in a separate repository on a separate release cadence; theories are built programmatically today, and nothing else in the crate depends on the syntax landing.
3. *Faithfulness.* Because the simulator and the interpreter share semantics but not code paths, their agreement is a permanent obligation. What runs today is a differential against an independently written reference matcher over a corpus of several hundred cases, together with a negative control establishing that the corpus can detect a broken matcher. The differential against the interpreter’s own matcher is written, is gated behind a build feature, and has not been run, because the workspace it needs does not currently build from a clean checkout. The gap is visible in the test output by design rather than recorded only here.
4. *The dilation backend.* The crate does not yet assemble the sparse complex matrix T_0 , compute a certified normalisation, form the defect square roots, or synthesize the auxiliary-qubit unitary. Until those pieces exist, §?? is a mathematical construction rather than an implemented execution path.

Two known incompletenesses. First, the partition check evaluates (P1) and (P2) against a supplied set of witnesses. It is therefore a sound *refuter* — an overlap it reports is real — and not a proof of admissibility, which would require deciding entailment in the generated logic. By Observation ?? this is exactly the price of taking the external branch, and it should be read as such rather than as a deficiency of the check. Second, the checkable-fragment gate currently admits modal and fixed-point formulae as keys, which ?? forbids; enforcing ?? at elaboration time is a small change and is the precondition for the incremental scheme below.

9.3 Incremental propensity

Lemma ?? licenses the standard optimisation. Maintain $\mathbf{a}$ as a Fenwick tree over (r, φ) classes with per-class redex lists; after firing at position k , reclassify only redexes within radius $\alpha + |L| + |R|$ of k , and additionally rescale the classes whose weight the update function touched — which for Definition ?? is exactly one class. The second part is the only respect in which dynamic weight maps cost anything at run time, and it is cheap precisely when update functions are sparse. A modeller who writes an update function touching every entry pays a full recomputation of $\mathbf{a}_0$ per step, and should be told so.

10 Weights and costs

Cost accounting and weighting both decorate rewrites, and it is worth stating plainly that they are different decorations doing different jobs.

A cost decoration attaches to a rewrite a debit against a token stack, and the funding judgment $\Sigma \vdash_{\S} \Delta$ — decidable, total, monotone in supply — decides whether a step may occur at all [?]. It is implemented in the transpiler branch as `OsIfResourceLogic`, with a supply multiset keyed by signature and a demand list, one token per signed term, and conformance laws asserting soundness, rejection of underfunding, supply monotonicity, and decidability. A weight decoration attaches a rate, and decides how fast and how often. The composition is one-directional:

Definition 60 (Funding gate). For a redex (k, σ) of rule r , let $\Delta(r, k, \sigma)$ be the demand it incurs and $\Sigma(P, k)$ the supply available at that location. Put $\chi(r, k, \sigma) = 1$ if $\Sigma(P, k) \vdash_{\S} \Delta(r, k, \sigma)$ and 0 otherwise.

Proposition 61 (Cost gates the support, weight distributes over it). *With χ as in Definition ??, the coalgebra of Definition ?? restricts to the funded sub-transition-system: unfunded redexes have propensity zero and contribute nothing to $\mathbf{a}_0$, while among funded redexes the distribution is unchanged from the cost-free case up to normalisation.*

Proof. χ appears as a multiplicative factor in Definition ??, taking values in $\{0, 1\}$; a zero factor removes the term from the sum, and the remaining terms are untouched. $\square$

Three consequences worth recording.

The gate is decidable, so the simulator stays exact. If funding were a real-valued penalty rather than a Boolean, the propensity would depend on an optimisation and the chain would no longer be simulable by Definition ?. The decidability law of the resource logic is doing real work here.

Exhaustion is a halting condition, not a deadlock. $\mathbf{a}_0 = 0$ when every enabled redex is unfunded, and Definition ?? halts. The chain has reached an absorbing state, which is the correct reading: a system that cannot pay for any of its available transitions has stopped, and the waiting time to the next event is infinite rather than undefined.

The neural example acquires a metabolism. Give the communication rule a phlogiston demand per firing and the network of §?? has an energy budget: a neuron whose local supply is exhausted has propensity zero regardless of its synaptic weights, and the network’s activity is bounded by its funding rather than by its capacity guard. Weight is what a synapse is worth; cost is what a spike costs. Separating them is what lets a model say that a strong synapse on a starving neuron is silent, which is a statement neither decoration makes alone. This is the point of contact with the mortal-computation reading of [?], and we go no further with it here.

11 Open threads

1. *A combined logic.* §?? supplies projectors for states conditioned on an all-zero auxiliary record. What one wants is a temporal logic that can speak both about generated configuration predicates and about success and failure histories, for example a modality for “after n successful applications of T ”. The conditioning event must be part of the semantics rather than silently renormalised away.
2. *Weighted bisimulation.* The category **GSLT** has bisimulation-preserving maps as morphisms [?]. Rate bisimulation is the natural real-valued candidate. In the complex case a relation must preserve the amplitude operator, not merely the measured populations: a plausible criterion is an intertwiner for T , but it is not yet clear whether the defect operators, or only the postselected completely positive map $\rho \mapsto T\rho T^\dagger$, must also be preserved.
3. *Derivation-sensitive phases.* Definition ?? sums over individual derivations, but all derivations in one refinement class read the same map entry. More fine-grained interference requires keys or decorations that distinguish derivations without making the map grow with the term. Whether that can be reconciled with Lemma ?? is open.
4. *Efficient dilation.* The canonical defect square roots are mathematically explicit and can be dense even when T_0 is sparse. A practical backend should exploit sparse block encodings, approximate polynomial square roots, or a larger but cheaper dilation while preserving the exact all-zero block. The resource trade-off between auxiliary dimension, circuit depth, and postselection probability has not been developed here.

5. *Geometric factors.* §?? introduces g as a real structural weight. Definition ?? treats it as an amplitude multiplier. A modeller who wants $|T_{c',c}|^2$ to retain a particular classical geometric scaling may instead want $\sqrt{g}$. The right choice is part of the quantum model and should not be hidden behind a supposed conservativity theorem.
6. *Quantale weights.* A quantale-valued codomain gives a fuzzy calculus, and [?] develops the flat-substitution instance independently. The criterion is no longer the existence of a map into $\mathbb{R}_{\geq 0}$: the real and complex cases already have different operational readings. The open question is instead which algebraic structure turns a weighted coalgebra into an executable object and what replaces unitary dilation outside $\mathbb{C}$.
7. *Infinite configuration spaces.* Remark ?? uses finiteness to make T_0 and its defects concrete. The construction extends to bounded contractions on a separable Hilbert space, but reachable complex weight maps can easily make the operator unbounded or its norm noncomputable. Useful syntactic conditions guaranteeing boundedness would remove the present quantisation requirement.
8. *Modal-key locality.* ?? excludes behavioural modalities from keys because Lemma ?? fails for them, and a weight that depends on what a redex can do next is a natural thing for a modeller to want. The question is whether a bounded-lookahead modality $\langle K \rangle^{\leq n}$ admits a locality lemma with radius depending on n , or whether modal keys simply cost global reclassification at every step.

References

- [1] L. G. Meredith. *Classifying knots with spatial-behavioural logics*. F1R3FLY.io working note, 2026.
- [2] L. G. Meredith. *Cost accounting as an endofunctor on continued interactive GSLTs*. F1R3FLY.io working note, 2026.
- [3] L. G. Meredith. *Cost, spacetime, and the geometry/matter split on reduction graphs*. F1R3FLY.io working notes, 2026.
- [4] *Fuzzy rho-calculus: a quantale-graded interaction, and its lift to ciGSLTs with binding*. F1R3FLY.io working note, 2026.
- [5] M. A. Gibson and J. Bruck. Efficient exact stochastic simulation of chemical systems with many species and many channels. *Journal of Physical Chemistry A* 104(9):1876–1889, 2000.
- [6] D. T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *Journal of Physical Chemistry* 81(25):2340–2361, 1977.
- [7] F1R3FLY.io. `mettail-gillespie`: stochastic and quantum Gillespie simulation extensions for MeTTaIL rewrite rules. F1R3FLY-io/publications, rho-life/Gillespie, 2026.
- [8] D. O. Hebb. *The Organization of Behavior*. Wiley, 1949.
- [9] L. G. Meredith. *Knots as cost-accounted processes*. F1R3FLY.io working note, 2026.
- [10] W. S. McCulloch and W. Pitts. A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics* 5:115–133, 1943.

- [11] L. G. Meredith et al. *The MeTTa calculus*. F1R3FLY.io, 2026.
- [12] F1R3FLY.io. `mettail-rust`: the `language!` macro and its shipped language definitions. F1R3FLY-io/mettail-rust, 2026.
- [13] L. G. Meredith and M. Radestock. Namespace logic: a logic for a reflective higher-order calculus. In *Trustworthy Global Computing*, LNCS 3705, Springer, 2005.
- [14] L. G. Meredith. *Graph-Structured Lambda Theories: A Reference for the Working Developer*. F1R3FLY.io, 2026.
- [15] C. Priami. Stochastic π -calculus. *The Computer Journal* 38(7):578–589, 1995.
- [16] L. G. Meredith. *The Mortal Scientist*. F1R3FLY.io working note, 2026.
- [17] A. Phillips and L. Cardelli. A correct abstract machine for the stochastic pi-calculus. In *Bioconcur*, 2004.
- [18] A. Phillips and L. Cardelli. Efficient, correct simulation of biological processes in the stochastic pi-calculus. In *Computational Methods in Systems Biology*, LNCS 4695, Springer, 2007.
- [19] G. Q. Bi and M. M. Poo. Synaptic modifications in cultured hippocampal neurons: dependence on spike timing, synaptic strength, and postsynaptic cell type. *Journal of Neuroscience* 18(24):10464–10472, 1998.